MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 50C5

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Miniature, beam power, pentode

Outline --- 5-3

Base --- E7-1

Envelope --- T5-1/2

Cathode --- Coated unipotential

Base connections:

Pin No.	1	2	3	4	5	6	7
Element	g3 k	g1	h	h	g1	g2	p

ABSOLUTE MAXIMUM RATINGS:

Parameter:	Ef	Eb	Ec1	Ec2	Pp	Pg2	Ehk	TE	Alt
Unit:	V	Vdc	Vdc	Vdc	W	W	v	°C	ft
Maximum:	55	150	---	130	6.0	1.4	100	230	(See note 2)
Minimum:	45	---	---	---	---	---	---	---	---
TEST CONDITIONS:	50	110	-7.5	110	---	---	---	---	---

GENERAL:

Qualification - Required

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:21Z
Check the source to verify that this is the current version before use.

METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS		UNIT
						Min	Max	
	Quality conformance inspection, part 1							
1266	Total grid current		0.65	II	Ic1	0	-3.0	µAdc
1256	Electrode current (anode)		0.65	II	Ib	54	67	mAdc
1341	Power output	Esig = 5.3 Vac Rp = 2,500 ohms	0.65	II	Po	1.4	---	W
1246	Audio frequency noise	Esig = 200 mVac Rp = 2,000 ohms	0.65	II	---	---	---	---
1231	Emission current	Eb = Ec1 = Ec2 = 30 Vdc	0.65	II	Is	150	---	mAdc
1201	Short and discontinuity detection	See note 1	0.4	II	---	---	---	---
	Quality conformance inspection, part 2							
1031	Low-frequency vibration	Ec1 = -13.5 Vdc Rp = 2,000 ohms	6.5	S3	Fp	---	1000	mVac
1301	Heater current		6.5	S3	If	138	162	mA
1336	Heater-cathode leakage		6.5	S3	Ihk	0	75	µAdc
1256	Electrode current (screen)		6.5	S3	Ic2	0.8	8.5	mAdc
1306	Transconductance		6.5	S3	Sm	6000	9500	umhos
1105	Permanence of marking		---	---	---	---	---	---
1211	Insulation of electrodes		6.5	S3	R	10	---	Meg
1121	Base strain	See note 3	---	---	---	---	---	---
2126	Glass strain		2.5	I	---	---	---	---
	Quality conformance inspection, part 3							
1501	Intermittent life test (500 hours)	Group A Ehk = 100 V Eb = Ec2 = 135 Vdc Rk = 215 ohms Rg1 = 0.5 meg	---	---	---	---	---	---
----	Life-test end points (500 hours)	Power output	---	---	Po	0.95	---	W

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:21Z
Check the source to verify that this is the current version before use.

NOTES:

1. Paragraph 2.2.2 of Method 1201 shall apply.
2. For operation at any altitude, the maximum peak voltage between any adjacent pins is limited in accordance with the rating 6.5.2.3. The equipment designer is cautioned to consider that voltage breakdown is only one factor influenced by altitude, and that other tube ratings especially temperature must also be observed. If operating voltages not exceeding 200 Vrms are used between adjacent pins, this type may be utilized at any altitude without derating for altitude effects.
3. Electrical rejects other than inoperatives may be used in the performance of this test.

Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Army - EL
Navy - EC
Air Force - 11, 17
DESC - ES

User activities:
Army - MU
Navy - WP, MC, CG
Air Force - None

Preparing activity:
Air Force - 80

(Project 5960-2525)

Source: https://assist.dla.mil -- Downloaded: 202610-15T13:21Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:21Z
Check the source to verify that this is the current version before use.

FOLD

POSTAGE AND FEES PAID
DEPARTMENT OF THE AIR FORCE

UNITED STATES AIR FORCE
OFFICIAL BUSINESS

SMAMA/MMREE
Wright-Patterson AFB OH 45433

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:21Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

   ☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

S/N-0102-014-1801 C-25254

Source: https://assist.dla.mil -- Downloaded: 202640-15T13:21Z
Check the source to verify that this is the current version before use.